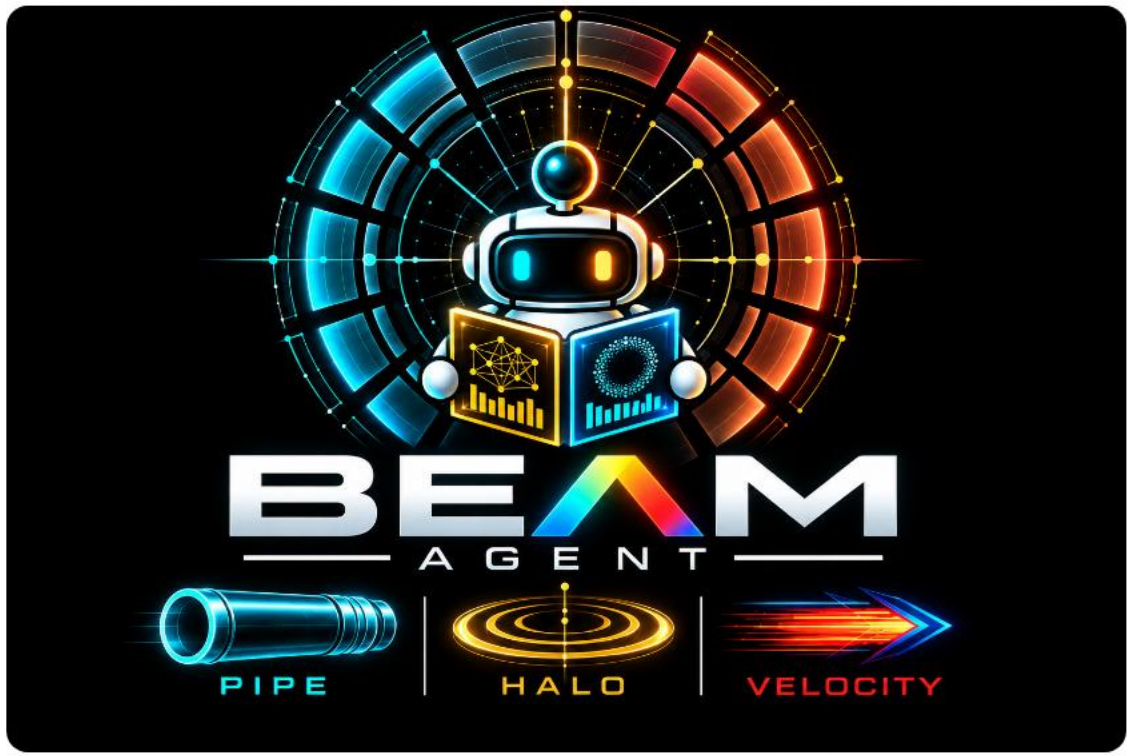


MK1 · IDEA AND PLAN

BEAM PIPE

Pipeline for Inference, Proposals and Evidence

BEAM HALO

Hypotheses from Associations and Latent Overlap

BEAM VELOCITY

Project velocity, folded into the agent design

One local archive of the ATLAS LLP record, with the agent built on it. Optional link to Claude.

TWO INSTRUMENTS ON BEAM

BEAM PIPE is the agent; BEAM HALO is a batch job

AGENT

BEAM PIPE

Responds to my request. It decides which tool to use, reads the result, chooses the next step, and loops until it can answer or says it cannot. It also proposes actions for my approval.

Decision-making at each step is what makes it an agent.

BATCH SYNTHESIS

BEAM HALO

Runs a fixed procedure on a schedule: cluster the embeddings, build the concept graph, sweep problems against texts, score, and queue the survivors.

Nothing in it decides what to do next. It is closer to an overnight analysis script.

Both run on BEAM, the shared local index. HALO is built as batch first; an agentic version comes only if its hits prove worth chasing.

THE IDEA

Turn a local library into dense, cited prompts and new hypotheses

1 Library

Papers, ML texts, my own record and code, kept in local folders.

2 Index

Each passage is chunked, tokenized and embedded as a vector, then tagged with source, date and status.

3 Query

I type a short question. The index returns the few passages that matter.

4 BEAM PIPE

The agent searches in a loop and assembles a dense, cited prompt.

5 Answer

A local model answers, or the prompt goes to Claude for the hard reasoning.

6 BEAM HALO

The same vectors are mined for links between sources, giving hypotheses to vet.

THE IDEA

One archive, two instruments built on it

Core · phases 1 to 4

BEAM

The archive: what is known, and where it is written

- Whole corpus searchable, offline
- Every answer quotes its passage
- Canon outranks superseded

On BEAM · phases 5 to 8

BEAM PIPE

The agent: what to look at next, and what to do about it

- Picks tools, searches in a loop
- Proposes changes for approval
- Escalates hard reasoning to Claude

On BEAM · phases 9 to 10

BEAM HALO

Synthesis: links at the edges of the record

- Maps sources that never cite each other
- Sweeps open problems against the texts
- Outputs hypotheses, not findings

Every write waits in a queue for my approval, and ATLAS-internal material never leaves the box.

THE PLAN

A smart agent with my working library at hand

Library
Papers, ML texts, my record, my code

→

Index
BEAM: tokenize, chunk and embed locally. The main boost.

→

Agent
BEAM PIPE picks what to read and builds the prompt

→

LLM
Local model, or Claude for hard reasoning

→

Me
Thinking and writing code, with cited answers

BEAM HALO Runs in the background on the same index: maps links between sources that never cite each other, sweeps open problems against the texts, and queues the hypotheses for me to vet.

What the agent can reach

- Papers and ML texts, indexed locally
- My record: plans, lock-ins, refuted results
- The code we are writing, re-indexed as it changes

What the LLM receives

- A short request expanded into dense context
- Verbatim passages, cited and status-labelled
- Internal material goes to the local model only

mk1: BEAM only, a handful of sources, all local

What to do

1. Pick 3 to 5 papers and one textbook chapter, about 100 to 150 pages
2. Write 10 to 15 questions with the expected file and page, plus 2 or 3 not in the corpus, before running anything
3. Download the embedding model and PDF converter once; offline after that
4. Ingest: extract text and LaTeX, chunk, embed, store with page numbers
5. Query page: type a question, see the top passages and their scores
6. Score the questions, read every miss, adjust chunking or extraction, rerun

Gates to pass before mk2

- **G1** Every file ingests, and every chunk traces to its file and page
- **G2** At least 80% of questions return the right passage in the top 3
- **G3** Out-of-corpus questions score below the threshold
- **G4** Equations in returned passages are readable against the original

> Not in mk1: no chat model, no agent, no BEAM HALO, no Claude link. It shows what it would hand an LLM.

BENEFITS AND DRAWBACKS

What this buys, and where it breaks

Buys

- **Dense prompts:** a short query becomes the few passages worth sending
- **Grounding:** answers come from quoted passages, not reconstruction
- **Status filter:** superseded results arrive labelled, or not at all
- **Confidentiality:** the tier filter decides what leaves the box
- **Persistence:** the index remembers across sessions
- **Reproducibility:** the passages sent are logged and rerunnable
- **Separate scoring:** retrieval is measured apart from answers
- **Cost and focus:** fewer tokens, less irrelevant context
- **Model independence:** the index outlives any one LLM
- **Reuse:** the same embeddings feed the HALO map

Breaks

- A retrieval miss gives a confident answer from an incomplete set
- Local summaries are lossy, so passages go verbatim
- Small local models fail at long arguments and tool loops
- Embedders are weak on equations
- Status tags are kept by hand

> The gain comes from local retrieval choosing the passages. What is sent is plain text; the LLM tokenizes it again on its side.

ARCHITECTURE

One index, two ways to answer



Local route, default

Ollama on the 4070 SUPER: one 8 to 14B chat model, one embedder. All tiers.

Claude route, off by default

Question plus retrieved chunks over the API. Internal chunks blocked; preview before send.

THE AGENT LOOP

It picks a tool, reads the result, and decides the next step

1. Tool choice

Search papers, search the record, open a code file, or search again with a refined query.

2. Multi-step loop

Continues until it can answer with citations, or says it cannot. Hard step cap.

3. Actions

Proposes a status change, drafts a sweep problem, or escalates to Claude.

- Every write is a proposal in a queue; nothing changes until I approve it
- Tool calls are schema-constrained, and the full trace is shown with the answer
- Escalation to Claude carries public-tier chunks only

> Risk: 8 to 14B local models are unreliable at tool loops. The loop is scored on the eval set before it is trusted.

CORPUS

Five collections, tagged by what may leave the box

Collection	Contents	Tier	When
Papers	LLP, DisCo, ABCD, TEC; 10 to 20 papers, 10 to 50 pages each	Public	First
ML texts	Author-posted free textbooks	Public	For the sweep
Own record	Lock-in files, plans, handoffs, refuted-hypothesis ledgers	To tag	After papers
Code	FORGE src/llp and related, chunked by function	To tag	After papers
ATLAS notes	Internal notes	Internal	Local only

Equations are kept as LaTeX: arXiv source where it exists, a math-aware PDF converter otherwise. Each chunk keeps its page number.

THE REAL DESIGN WORK

Plain retrieval serves a retracted result with full confidence

So every chunk carries source, date, collection, tier and a status, and retrieval prefers canon.

canon

Currently pinned. Ranked first.

superseded

Replaced by a later result. Shown with a pointer to it.

refuted

Tested and wrong. Kept, labelled as such.

Example: λ=2, μ=1 as the endcap operating point is superseded; it fails the dCor gate. Proposed: status assigned by hand in the manifest.

PLAN

Eleven phases; the retrieval eval is the gate

#	Phase	What it produces
1	Gather and catalogue	Folders plus a manifest row for every source
2	Ingest and schema	Chunks with metadata; incremental re-index
3	Retrieval eval	20 to 30 known-answer questions, scored before any LLM
4	Local generation	Answers from retrieved chunks only, cited
5	Agent loop	Tool choice and multi-step search, with a trace; scored on the eval
6	Flask and front end	Plain page first; styling deferred
7	Claude route	API route with send-preview and internal block
8	Proposal queue	Status changes, sweep drafts and escalations awaiting approval
9	Latent map	Embedding clusters, concept graph, cross-source bridges
10	Sweep	Open problems run against textbook sections, bridges first
11	Optional LoRA	Only if the eval shows a gap retrieval cannot close

SYNTHESIS

Find links between sources that never cite each other

Embedding map

Cluster every chunk. Flag close neighbours that come from different fields.

Concept graph

Extract problem, method, assumption and math object per chunk. Look for A to B and B to C with no A to C.

Latent axes

Factorise the embeddings into a few dozen labelled directions shared across sources.

> Gate: it must rediscover links already known, such as DisCo and the distance-correlation statistics literature, before any new link is believed.

Limit: embedders are weak on equations, so the latent structure will track vocabulary more than mathematics. Every link is a hypothesis for vetting.

THE SWEEP

Brute force is how it finds what a human would not pair up

1

Write each open problem as a short statement

2

Run every problem against every textbook section, overnight

3

Score and keep the top few percent

4

Survivors go to Claude for hard vetting, then to me

Estimated GPU time: about an hour with a top-50 pre-filter per problem, about 50 hours for the full cross product. Most hits will be junk; the step-3 filter is the piece that needs designing.

TRAINING

Weight training is deferred, with the adapter slot kept open

Approach	Call at this corpus size
Retrieval	Build first. Updating means dropping in files and re-indexing.
LoRA on raw text	Skip. Too little text; shifts vocabulary, teaches no reliable facts.
Embedder fine-tune	Skip. Keyword plus vector search already handles the jargon.
RAFT-style LoRA	Feasible on 12 GB. Only if the eval shows a generation gap.

The RAFT route needs generated question and passage pairs. Whether that is acceptable is an open decision.

TIME

Rough estimates, in working hours

6–10 h

Cited chat over the papers: one to two evenings

40–55 h

Everything through the sweep: two to three weeks of evenings

What will blow the estimate

- PDF parsing of equations and two-column layouts
- Front-end polish beyond a plain page
- The agent loop and the latent map, both unproven

STEP 1

Gather the sources and catalogue them

Folders

```
0_papers/
0_ml_texts/
0_own_record/
0_code/
manifest.csv
```

Manifest row

- id, title, authors, year
- collection; tier (public or internal)
- format (tex or pdf); origin; local path
- status; date added

> Done when every file has a manifest row and every row's path resolves. A check script confirms it.

OPEN DECISIONS

Six calls still to make

Decision	Current proposal
Runtime	Ollama, unless llama.cpp directly is preferred
First slice	Papers, then own record and FORGE code
Status assignment	Hand-edited manifest
Generated training pairs	Needed only for RAFT; acceptable or not
Paper list	Needed to build the manifest and look up arXiv source
Escalation	Automatic for public chunks, or always ask first